# Networking Fundamentals

REGIONAL SCHOOL OF DEVOPS PROGRAM

The Network Team

# Agenda

1. NETWORKING GLOSSARY
2. NETWORK COMPONENTS AND MEDIA TYPES
3. NETWORK LAYERS AND MODELS
4. NETWORK SERVICES AND ROLES
5. TROUBLESHOOTING APPROACH AND TOOLS
6. RESOURCES

1

THIS WILL HELP US SOUND COOL AND GET THE MANAGERS TO LEAVE US ALONE

# Networking Glossary

**CONNECTION**

In networking, a connection refers to pieces of related information that are transferred through a network. This generally infers that a connection is built before the data transfer (by following the procedures laid out in a protocol) and then is deconstructed at the at the end of the data transfer.

**PORT**

A port is an address on a single machine that can be tied to a specific piece of software. It is not a physical interface or location, but it allows your server to be able to communicate using more than one application.

**PACKET**

A packet is the most basic unit that is transferred over a network. When communicating over a network, packets are the envelopes that carry your data (in pieces) from one end point to the other. Packets have a header portion that contains information about the packet including the source and destination, timestamps, network hops, etc. The main portion of a packet contains the actual data being transferred. It is sometimes called the body or the payload.

**NETWORK INTERFACE**

A network interface can refer to any kind of software interface to networking hardware. For instance, if you have two network cards in your computer, you can control and configure each network interface associated with them individually.

A network interface may be associated with a physical device, or it may be a representation of a virtual interface. The "loopback" device, which is a virtual interface to the local machine, is an example of this.

**NETWORK TOPOLOGY**

Various configurations, called topologies, have been used to administer LANs

**LAN**

LAN stands for "local area network". It refers to a network or a portion of a network that is not publicly accessible to the greater internet. A home or office network is an example of a LAN.

**WAN**

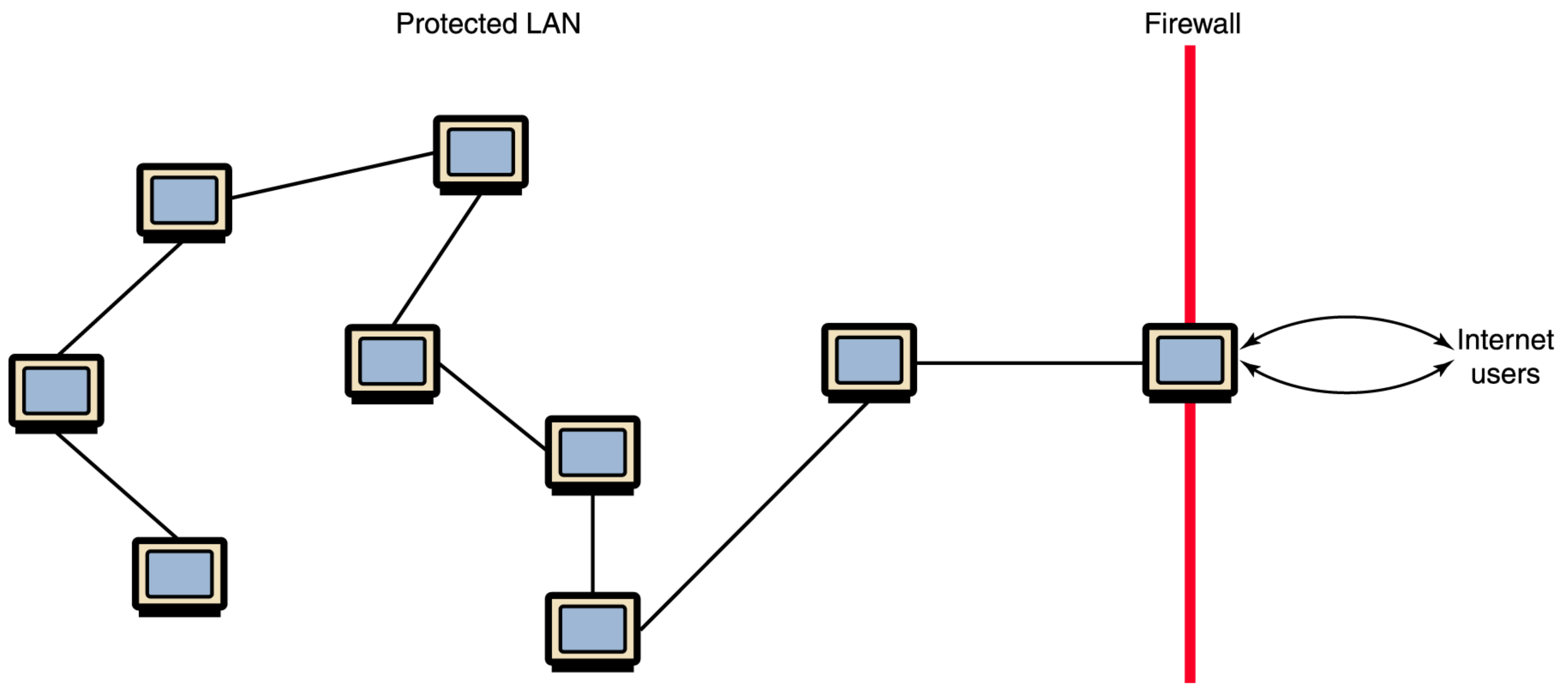
WAN stands for "wide area network". It means a network that is much more extensive than a LAN. While WAN is the relevant term to use to describe large, dispersed networks in general, it is usually meant to mean the internet, as a whole.

**NAT**

NAT stands for network address translation. It is a way to translate requests that are incoming into a routing server to the relevant devices or servers that it knows about in the LAN. This is usually implemented in physical LANs as a way to route requests through one IP address to the necessary backend servers.

**FIREWALL**

A firewall is a program that decides whether traffic coming into a server or going out should be allowed. A firewall usually works by creating rules for which type of traffic is acceptable on which ports. Generally, firewalls block ports that are not used by a specific application on a server.

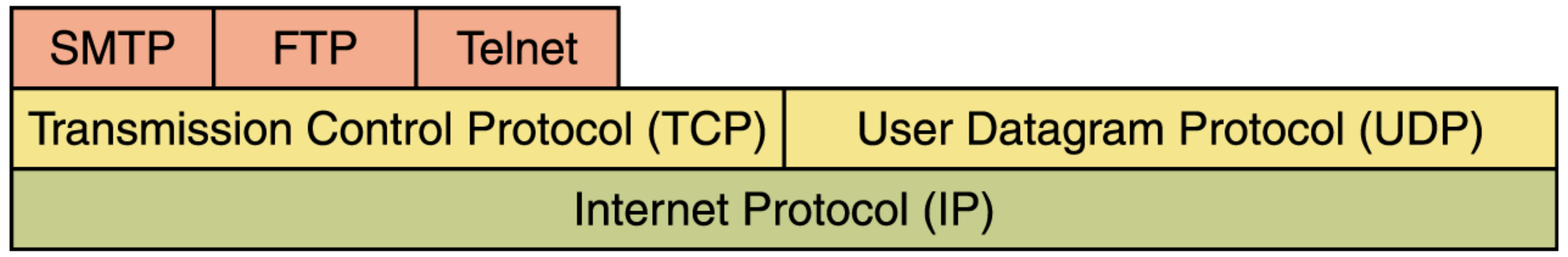



**VPN**

VPN stands for virtual private network. It is a means of connecting separate LANs through the internet, while maintaining privacy. This is used as a means of connecting remote systems as if they were on a local network, often for security reasons.

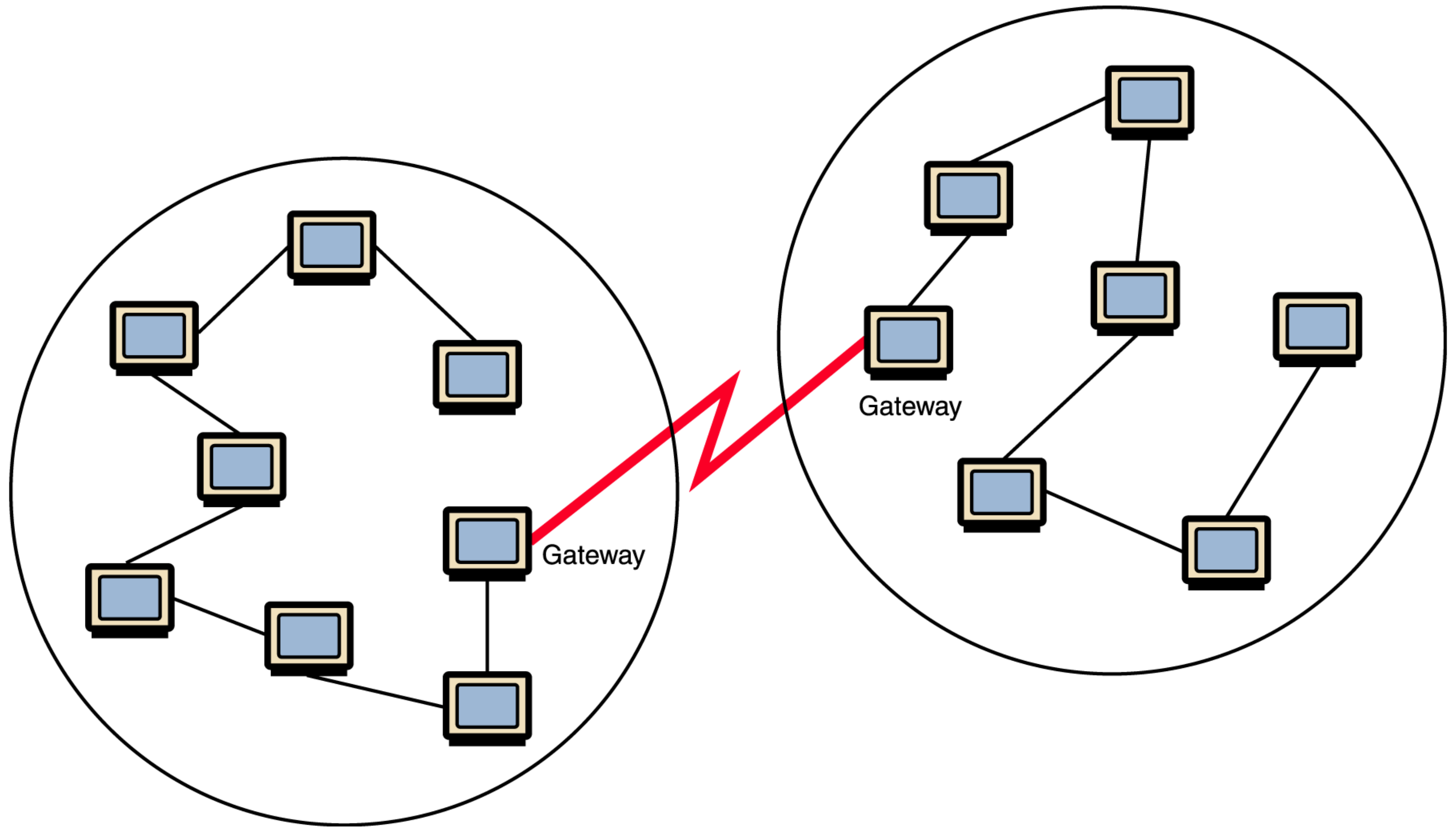
**PROTOCOL**

A protocol is a set of rules and standards that basically define a language that devices can use to communicate. There are a great number of protocols in use extensively in networking, and they are often implemented in different layers.

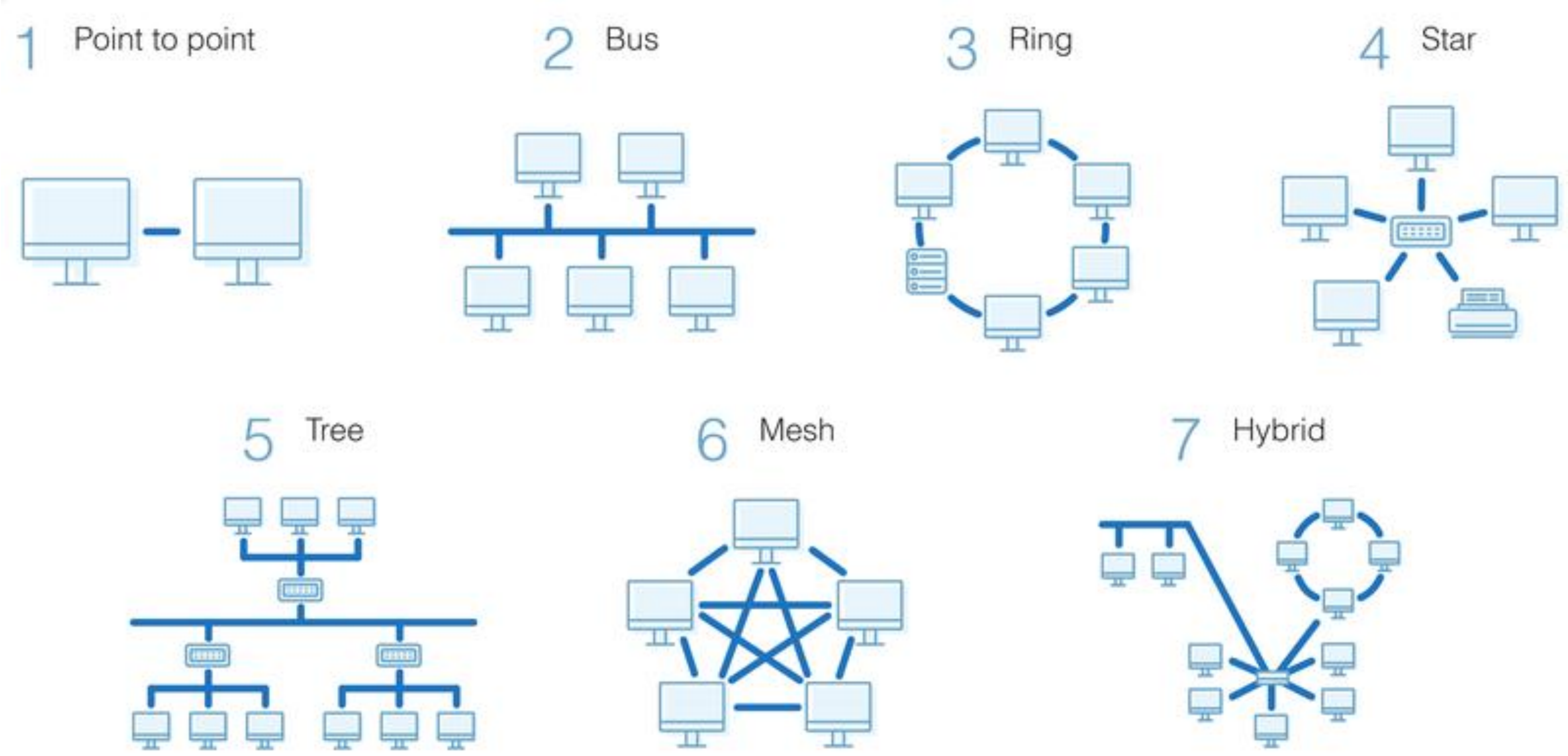


**GATEWAY**

A gateway is a piece of networking hardware or software used in telecommunications for telecommunications networks that allows data to flow from one discrete network to another. Gateways are distinct from routers or switches in that they communicate using more than one protocol to connect multiple networks




**TOPOLOGIES**



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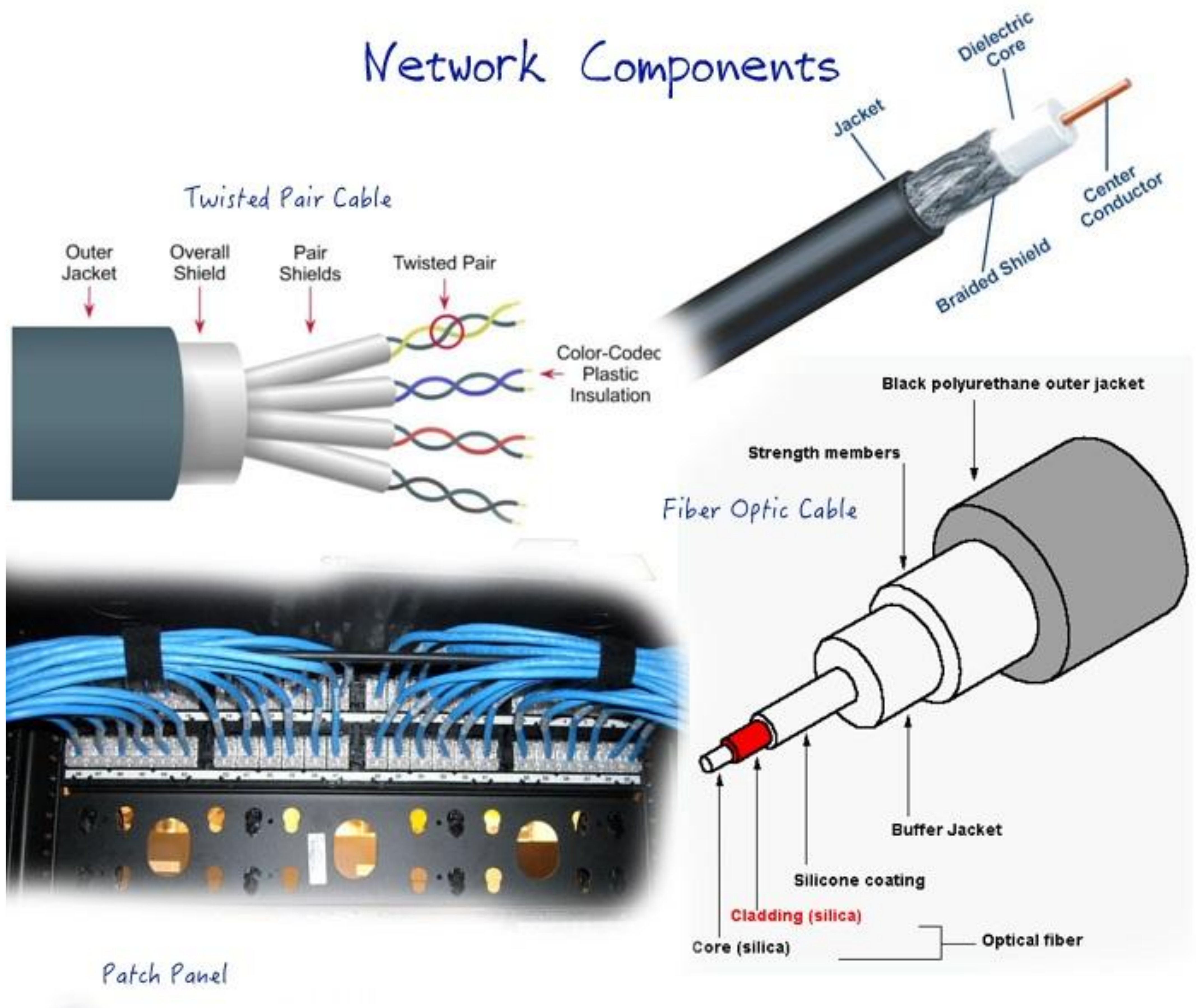
THERE IS A HIGHER CHANCE OF SEEING A LIVE DINOSAUR THAN SOME OF THESE THINGS

# Network components and media types





Switch

## Network Components

**Passive Network Components**

- Cable
- Junction box
- Connector
- Patch Panel
- Network Cabinet / Patch cabinet

**Active Network Components**

- Network card
- Repeater
- Hub
- Switch
- Router
- Gateway
- Server
- Proxy
- Print Server
- NAS (Storage)
- Firewall
- Load Balancer

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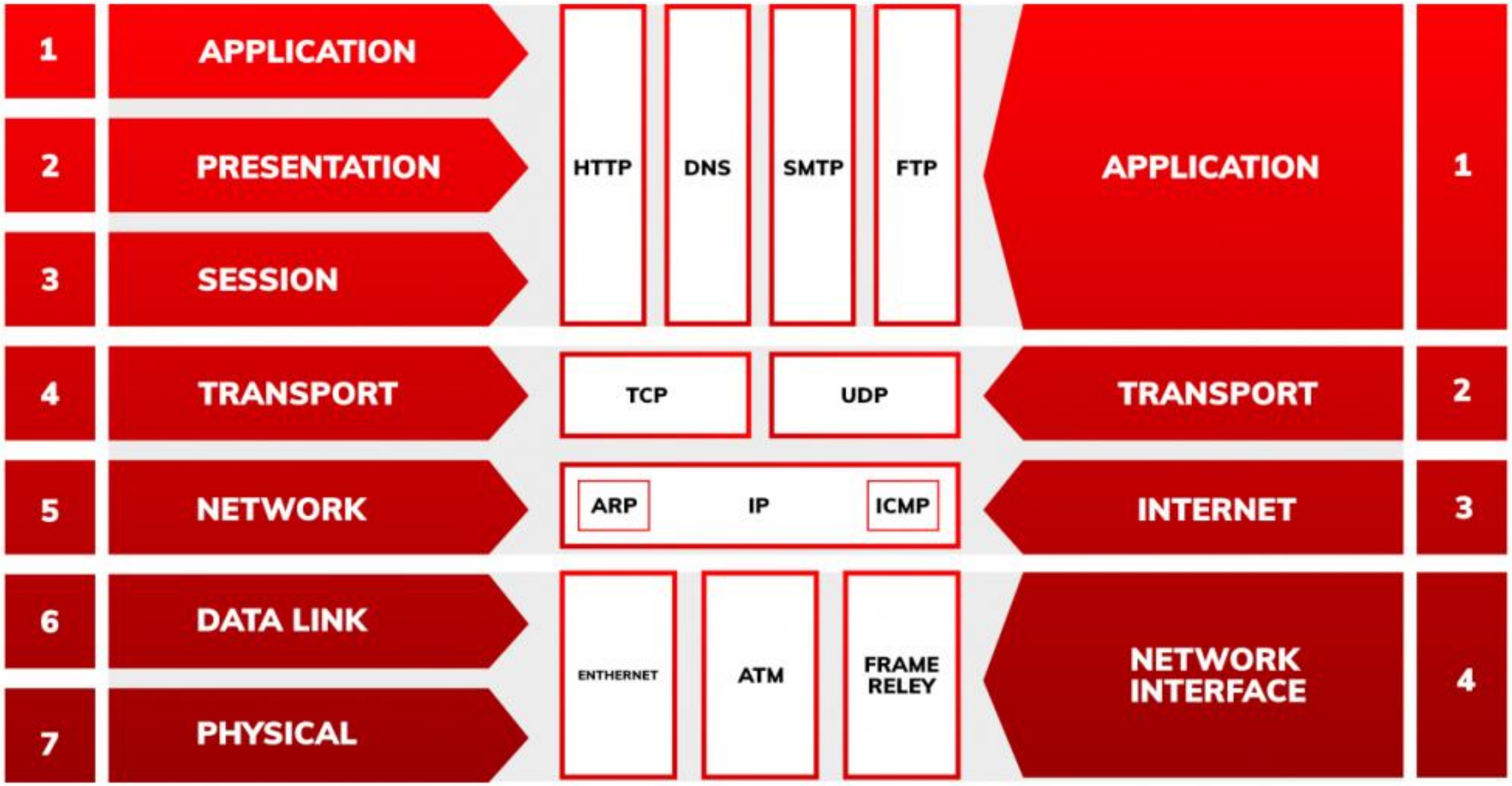
IF ANYTHING MADE SENSE TILL NOW FROM THIS POINT FEEL FREE TO LOSE ALL HOPE

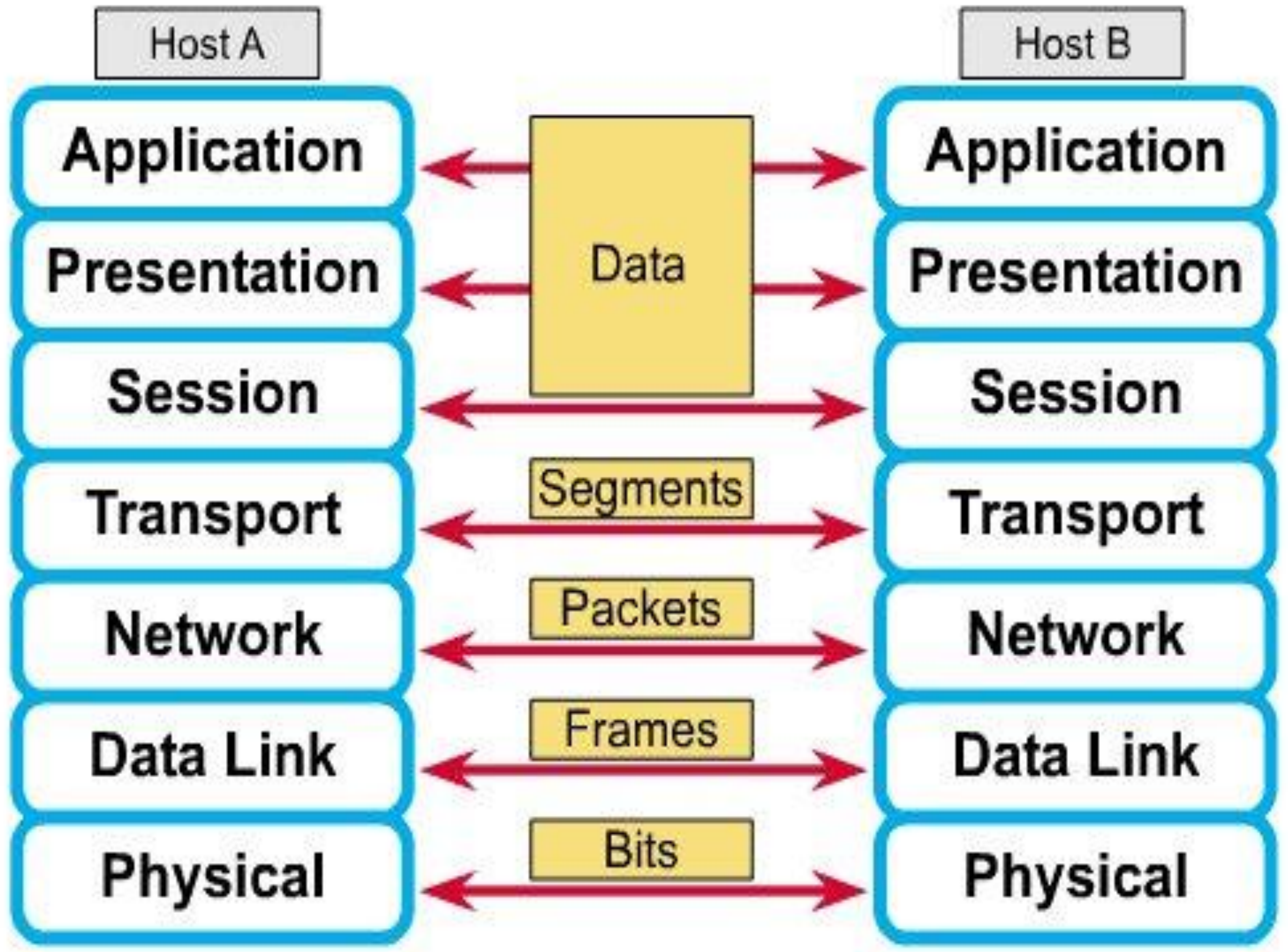
# Network Layers and Models

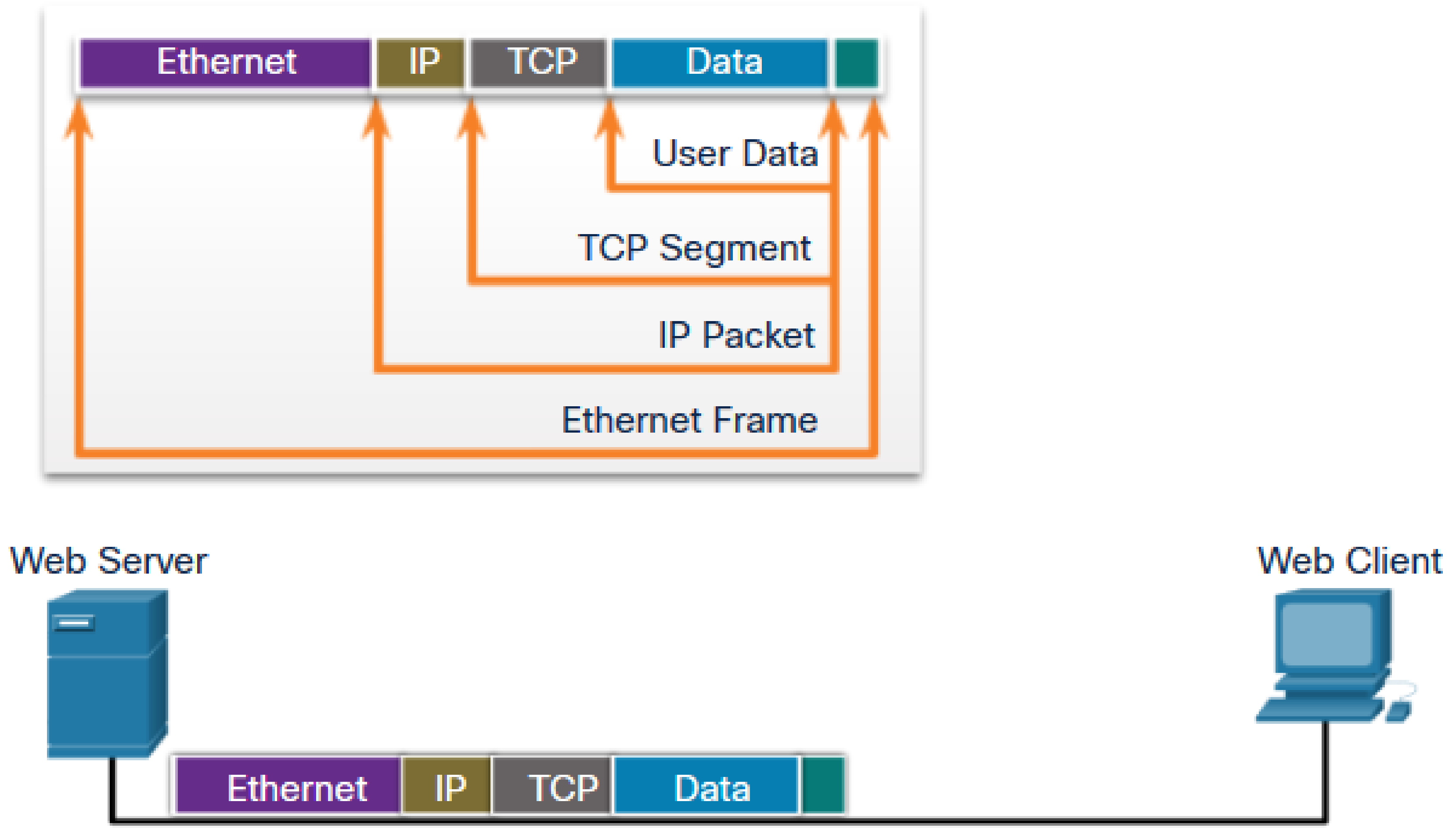
# Why?

# OSI vs TCP/IP

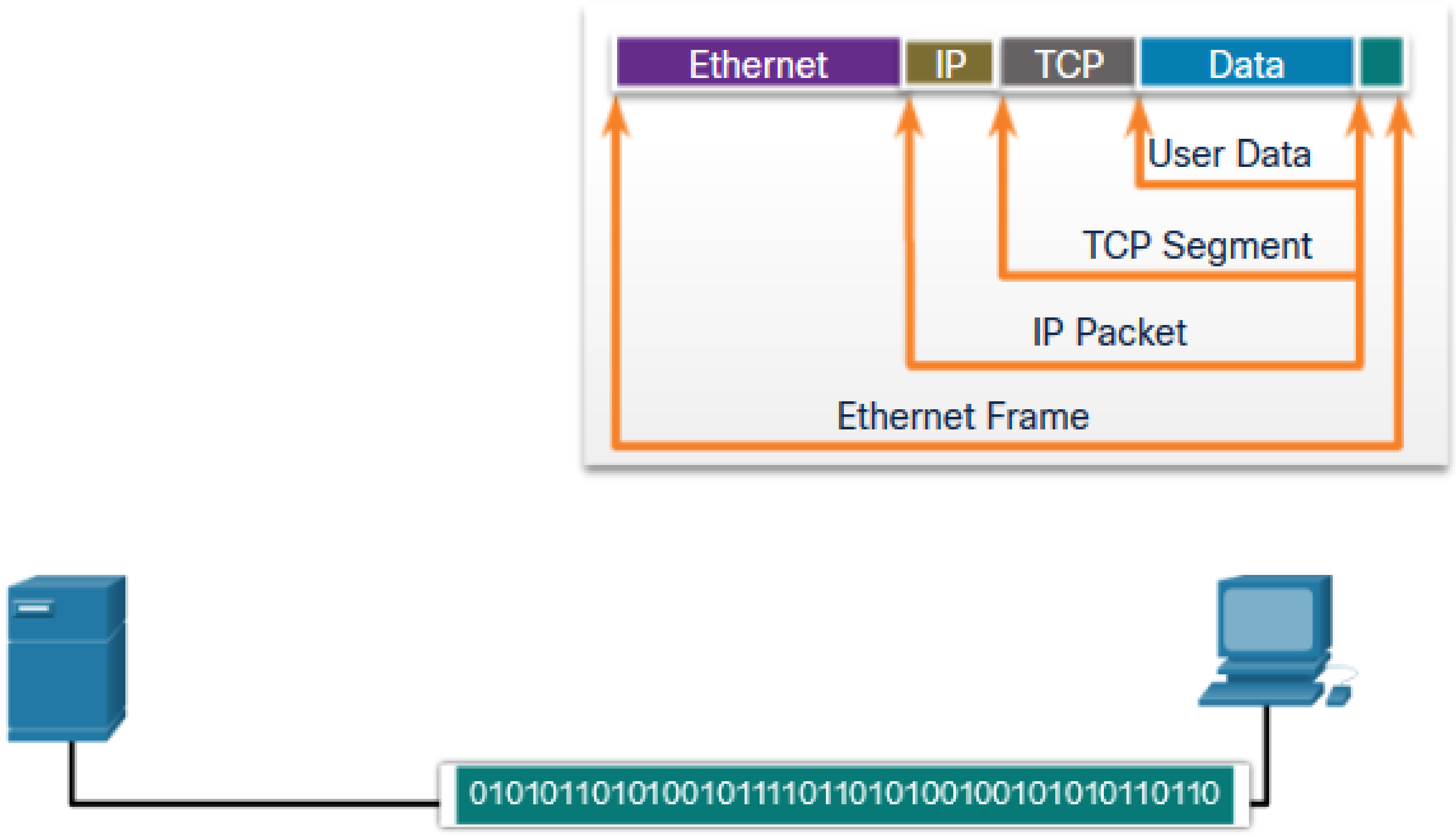

1
APPLICATION
2
PRESENTATION
3
SESSION
4
TRANSPORT
5
NETWORK
6
DATA LINK
7
PHYSICAL
HTTP
DNS
SMTP
FTP
TCP
UDP
ARP
IP
ICMP
ENTHERNET
ATM
FRAME RELEY
APPLICATION
1
TRANSPORT
2
INTERNET
3
NETWORK INTERFACE
4

# End to end connectivity

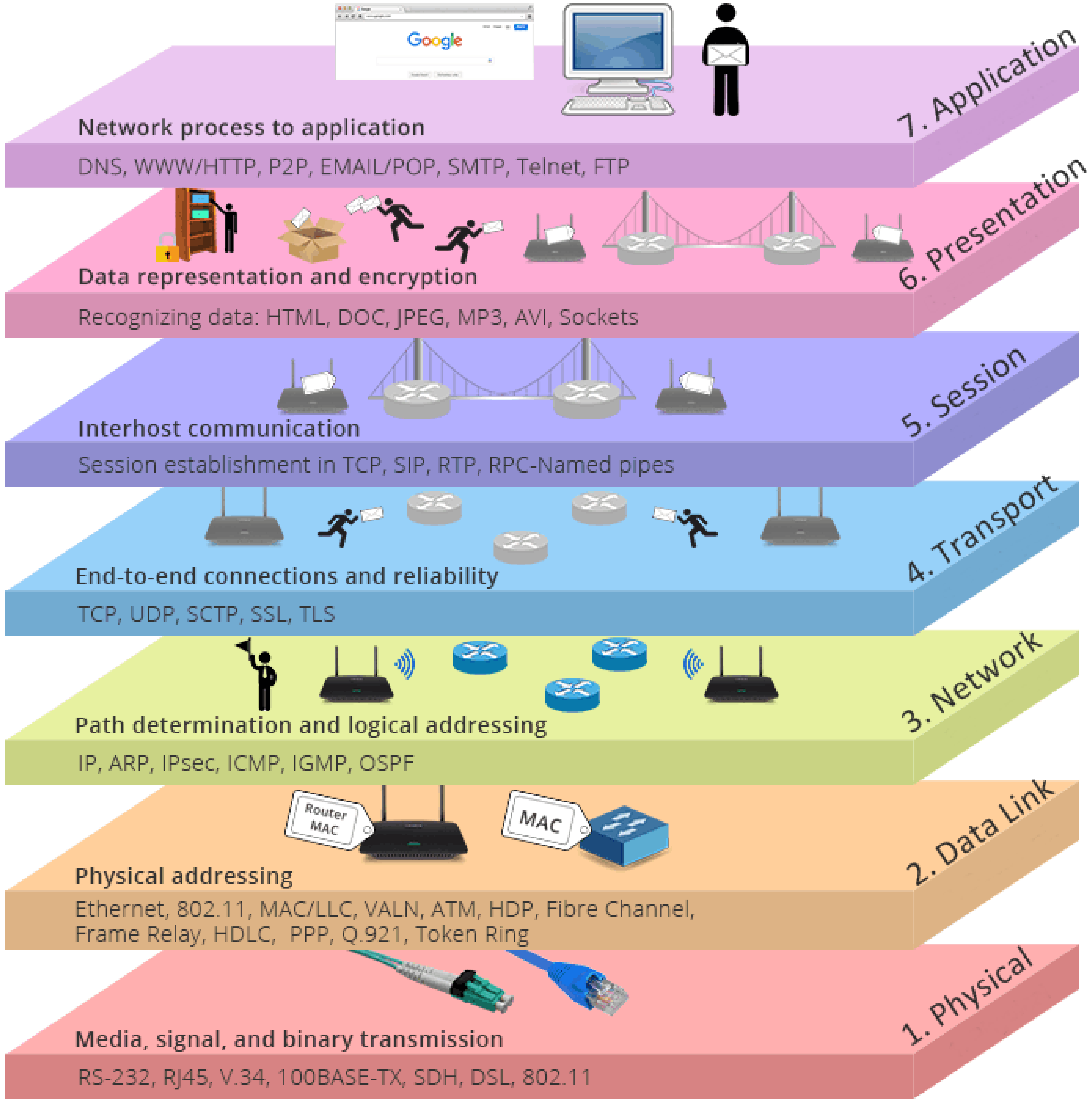


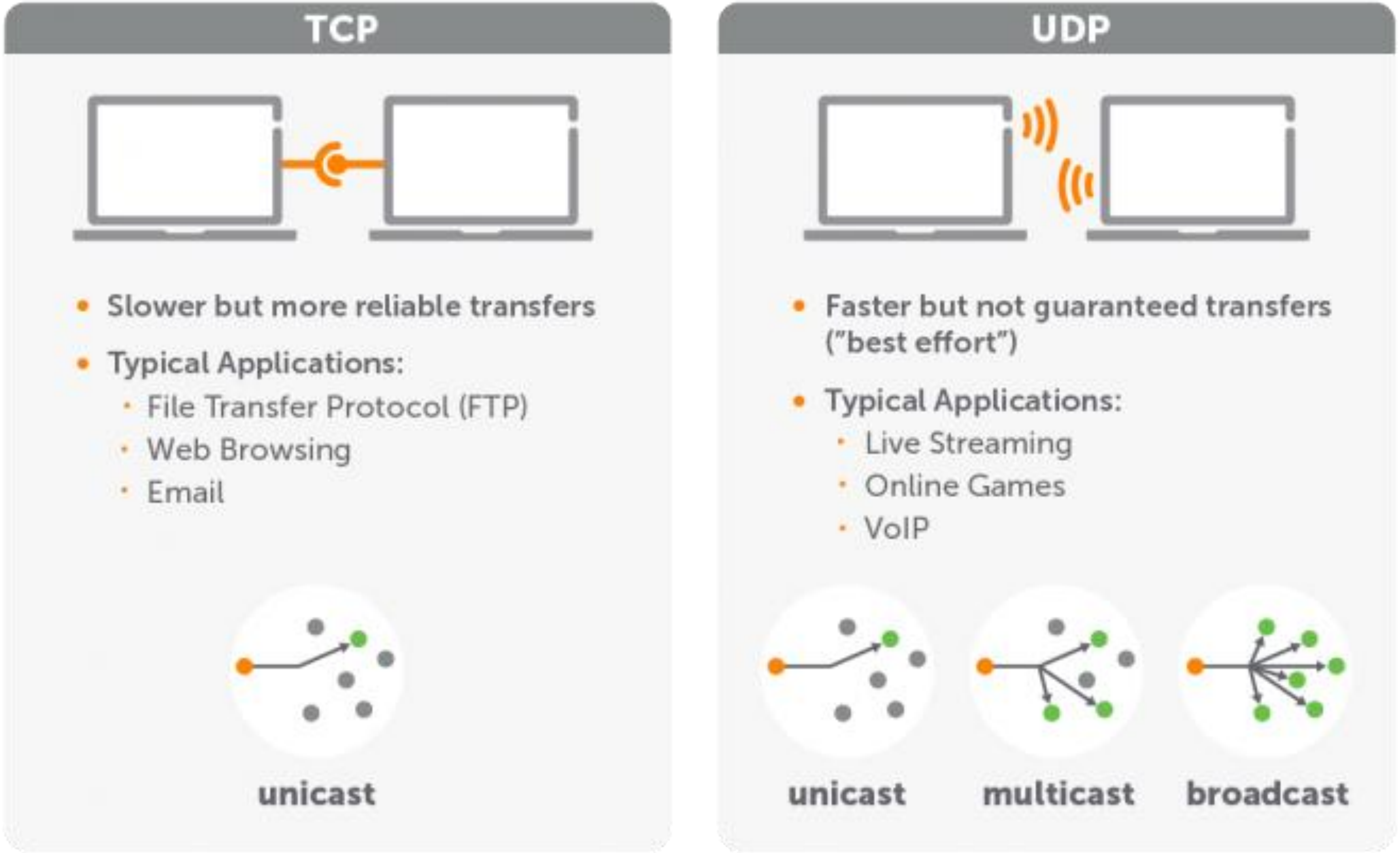

- **Each layer contains a Protocol Data Unit (PDU)**
  - PDU's are used for peer-to-peer contact between corresponding layers.
  - Data is handled by the top three layers, then Segmented by the Transport layer.
  - The Network layer places it into packets and the Data Link frames the packets for transmission.
  - Physical layer converts it to bits and sends it out over the media.
  - The receiving computer reverses the process using the information contained in the PDU.



# The OSI Model



# Transport Layer



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IF YOU EVER GOT AN A- AT A MATH TEST, PLEASE EXIT THE PRESENTATION QUIETLY...

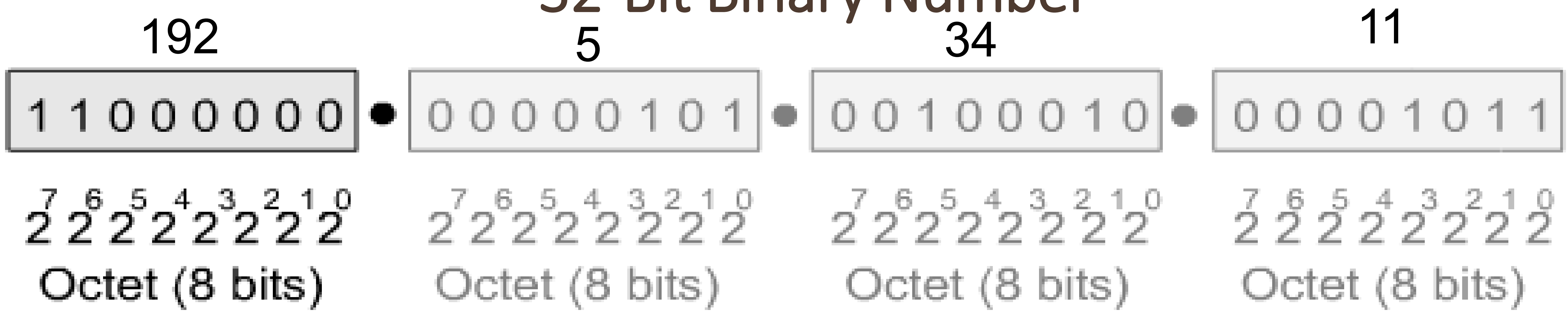
# Network Layer (IP)

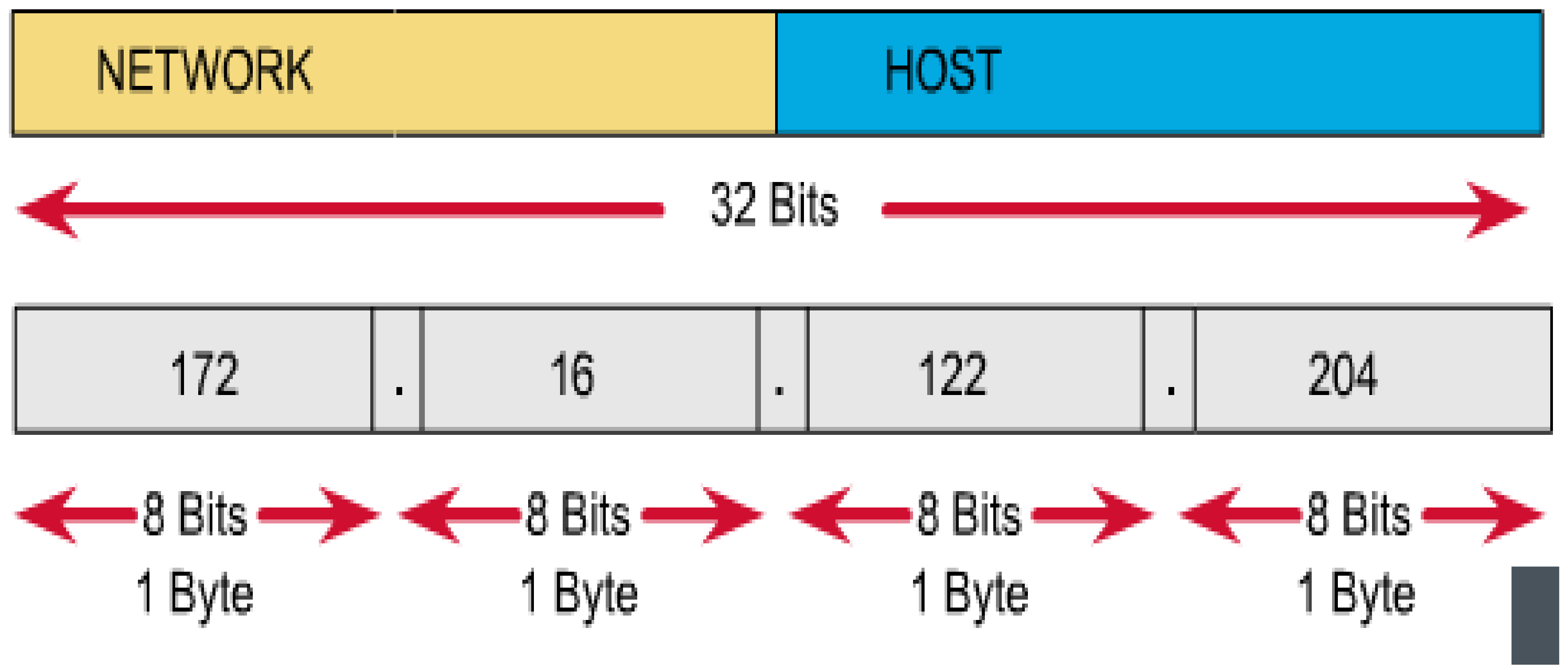
# Network Layer

| IPv4 | IPv6 |
| --- | --- |
| Deployed 1981 | Deployed 1998 |
| 32-bit IP address | 128-bit IP address |
| 4.3 billion addresses<br>Addresses must be reused and masked | $7.9 \times 10^{28}$ addresses<br>Every device can have a unique address |
| Numeric dot-decimal notation<br>192.168.5.18 | Alphanumeric hexadecimal notation<br>50b2:6400:0000:0000:6c3a:b17d:0000:10a9<br>(Simplified - 50b2:6400::6c3a:b17d:0:10a9) |
| DHCP or manual configuration | Supports autoconfiguration |

# Network Layer

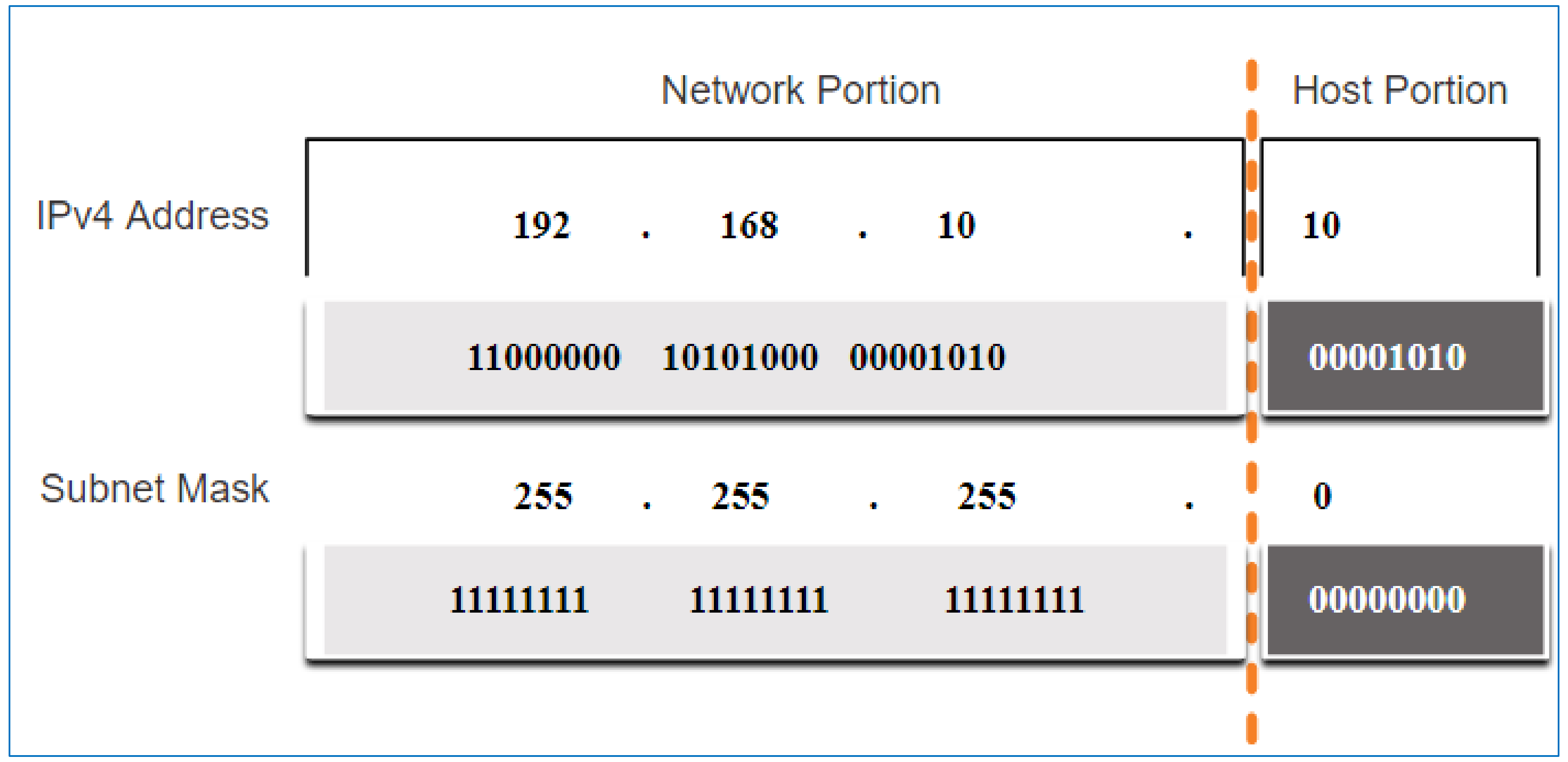
## IP Address as a 32-Bit Binary Number




- IP addresses are logical addresses (not physical)
- 32 bits.
- Includes a network ID and a host ID.
- Every host must have a unique IP address.
- IP addresses are assigned by a central authority (RIPE).

## Subnet Mask




MASK -determines which part of an IP address is the network field and which part is the host field

# Network Layer

| Class | 1st Octet Decimal Range | 1st Octet High Order Bits | Network/Host ID (N=Network, H=Host) | Default Subnet Mask | Number of Networks | Hosts per Network (Usable Addresses) |
|---|---|---|---|---|---|---|
| A | 1 – 126* | 0 | N.H.H.H | 255.0.0.0 | 126 ($2^7$ – 2) | 16,777,214 ($2^{24}$ – 2) |
| B | 128 – 191 | 10 | N.N.H.H | 255.255.0.0 | 16,382 ($2^{14}$ – 2) | 65,534 ($2^{16}$ – 2) |
| C | 192 – 223 | 110 | N.N.N.H | 255.255.255.0 | 2,097,150 ($2^{21}$ – 2) | 254 ($2^8$ – 2) |
| D | 224 – 239 | 1110 | Reserved for Multicasting | | | |
| E | 240 – 254 | 1111 | Experimental; used for research | | | |

**Note:** Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback and diagnostic functions.

## Private IP Addresses

| Class | Private Networks | Subnet Mask | Address Range |
|---|---|---|---|
| A | 10.0.0.0 | 255.0.0.0 | 10.0.0.0 - 10.255.255.255 |
| B | 172.16.0.0 - 172.31.0.0 | 255.240.0.0 | 172.16.0.0 - 172.31.255.255 |
| C | 192.168.0.0 | 255.255.0.0 | 192.168.0.0 - 192.168.255.255 |

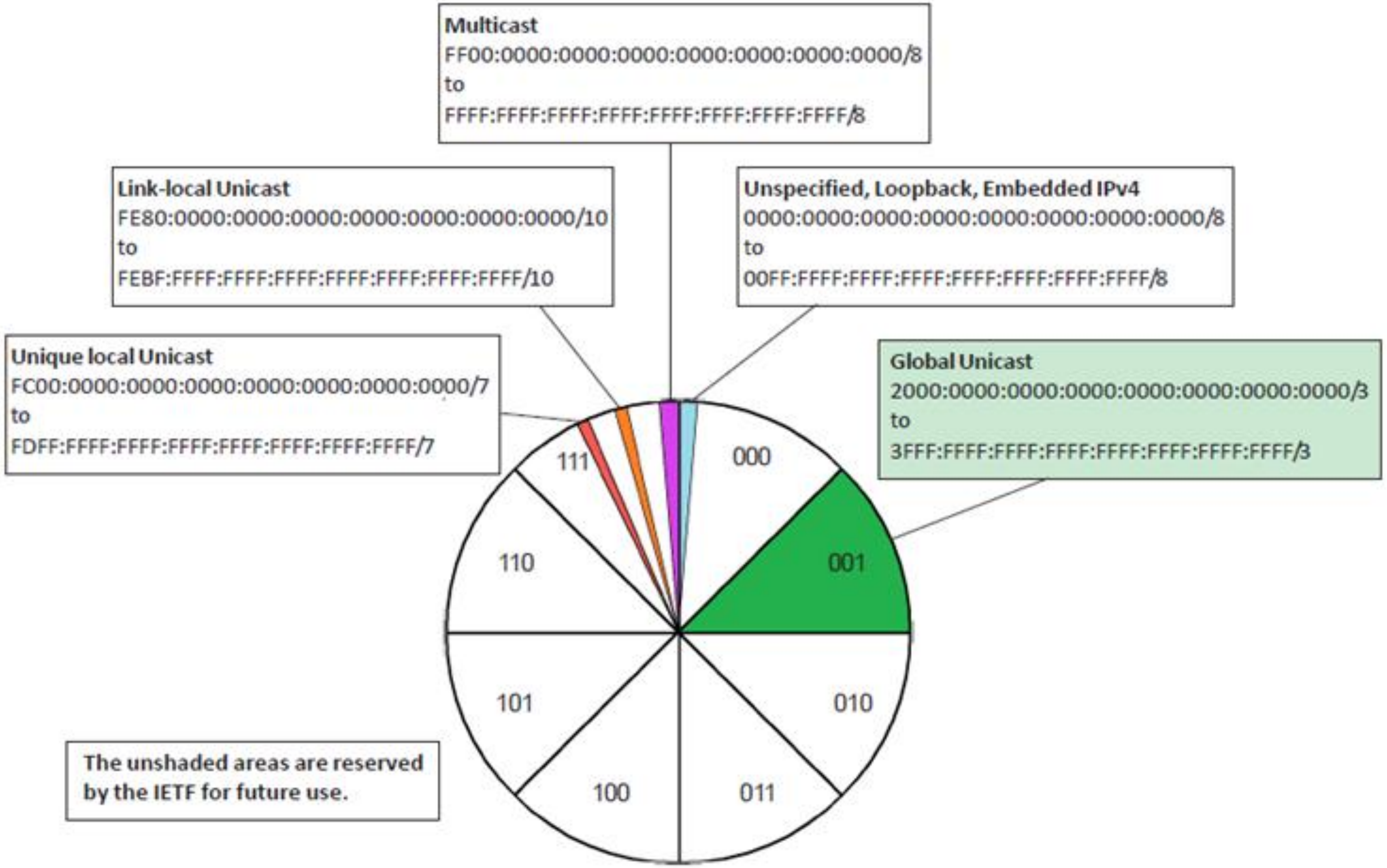
# Network Layer



| | | | Subnets | | | Valid Hosts | | |
|---|---|---|---|---|---|---|---|---|
| | | Subnet Mask | Class A | Class B | Class C | Class A | Class B | Class C |
| Class A | /8 | 255.0.0.0 | 1 | | | 16777214 | | |
| Class A | /9 | 255.128.0.0 | 2 | | | 8388606 | | |
| Class A | /10 | 255.192.0.0 | 4 | | | 4194302 | | |
| Class A | /11 | 255.224.0.0 | 8 | | | 2097150 | | |
| Class A | /12 | 255.240.0.0 | 16 | | | 1048574 | | |
| Class A | /13 | 255.248.0.0 | 32 | | | 524286 | | |
| Class A | /14 | 255.252.0.0 | 64 | | | 262142 | | |
| Class A | /15 | 255.254.0.0 | 128 | | | 131070 | | |
| Class B | /16 | 255.255.0.0 | 256 | 1 | | 65534 | 65534 | |
| Class B | /17 | 255.255.128.0 | 512 | 2 | | 32766 | 32766 | |
| Class B | /18 | 255.255.192.0 | 1024 | 4 | | 16382 | 16382 | |
| Class B | /19 | 255.255.224.0 | 2046 | 8 | | 8190 | 8190 | |
| Class B | /20 | 255.255.240.0 | 4096 | 16 | | 4094 | 4094 | |
| Class B | /21 | 255.255.248.0 | 8192 | 32 | | 2046 | 2046 | |
| Class B | /22 | 255.255.252.0 | 16384 | 64 | | 1022 | 1022 | |
| Class B | /23 | 255.255.254.0 | 32768 | 128 | | 510 | 510 | |
| Class C | /24 | 255.255.255.0 | 65536 | 256 | 1 | 254 | 254 | 254 |
| Class C | /25 | 255.255.255.128 | 131072 | 512 | 2 | 126 | 126 | 126 |
| Class C | /26 | 255.255.255.192 | 262144 | 1024 | 4 | 62 | 62 | 62 |
| Class C | /27 | 255.255.255.224 | 524288 | 2046 | 8 | 30 | 30 | 30 |
| Class C | /28 | 255.255.255.240 | 1048576 | 4096 | 16 | 14 | 14 | 14 |
| Class C | /29 | 255.255.255.248 | 2097152 | 8192 | 32 | 6 | 6 | 6 |
| Class C | /30 | 255.255.255.252 | 4194304 | 163384 | 64 | 2 | 2 | 2 |

https://www.calculator.net/ip-subnet-calculator.html

# IPV6 current distribution



# IPV6 representations

Rule 1: Leading 0s (zeroes) in any 16-bit section can be omitted.

Address **before** omission:
2001:0DB8:0001:5270:0127:00AB:CAFE:0E1F /64

Address **after** omission:
2001: DB8: 1:5270: 127: AB:CAFE: E1F /64

Rule2: use Double colon

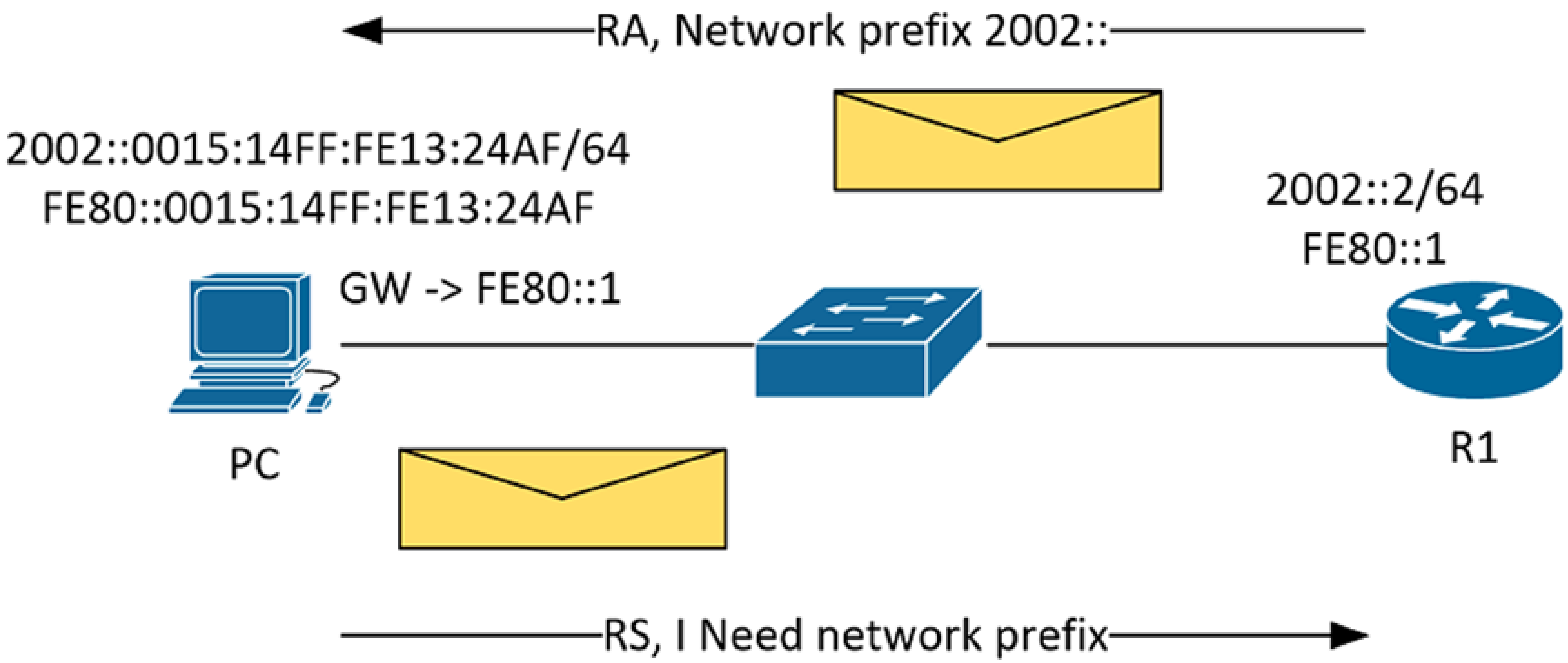
2001:0DB8:0000:0000:ACAD:0000:0000:E175

→ 2001:DB8::ACAD:0:0:E175

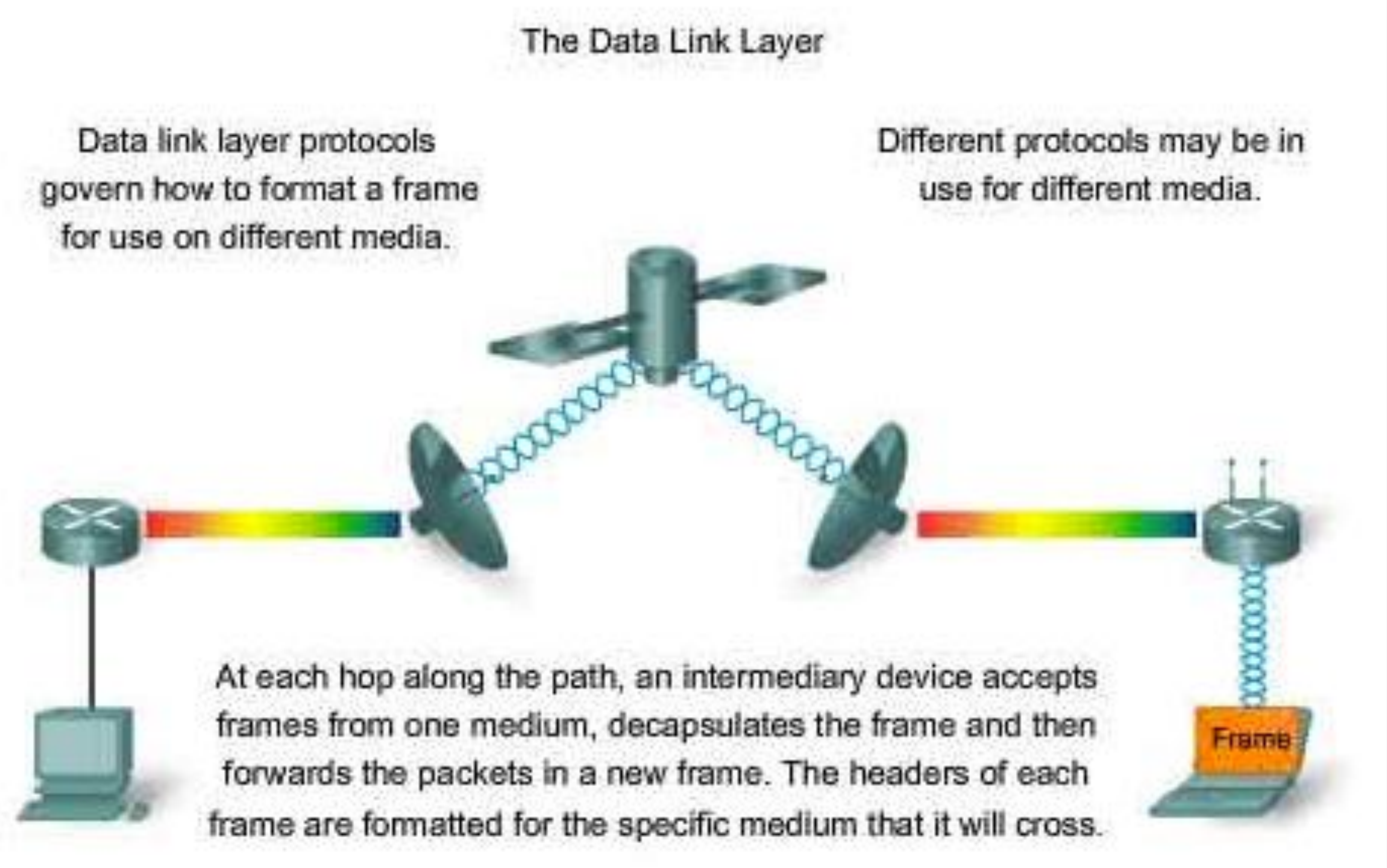
# IPV6 subnets


•The **Site Prefix or Global Routing Prefix** is the first 3 hextets or 48-bits of the address. It is assigned by the service provider.
•The **Site Topology or Subnet ID** Is the $4^{th}$ hextet of the address.
•The **Interface ID** is the last 4 hextets or 64-bits of the address. It can be manually or dynamically assigned using the EUI-64 command. (Extended Unique Identifier

# IPV6 SLAAC


RA, Network prefix 2002::
2002::0015:14FF:FE13:24AF/64
FE80::0015:14FF:FE13:24AF
GW -> FE80::1
2002::2/64
FE80::1
PC
R1
RS, I Need network prefix

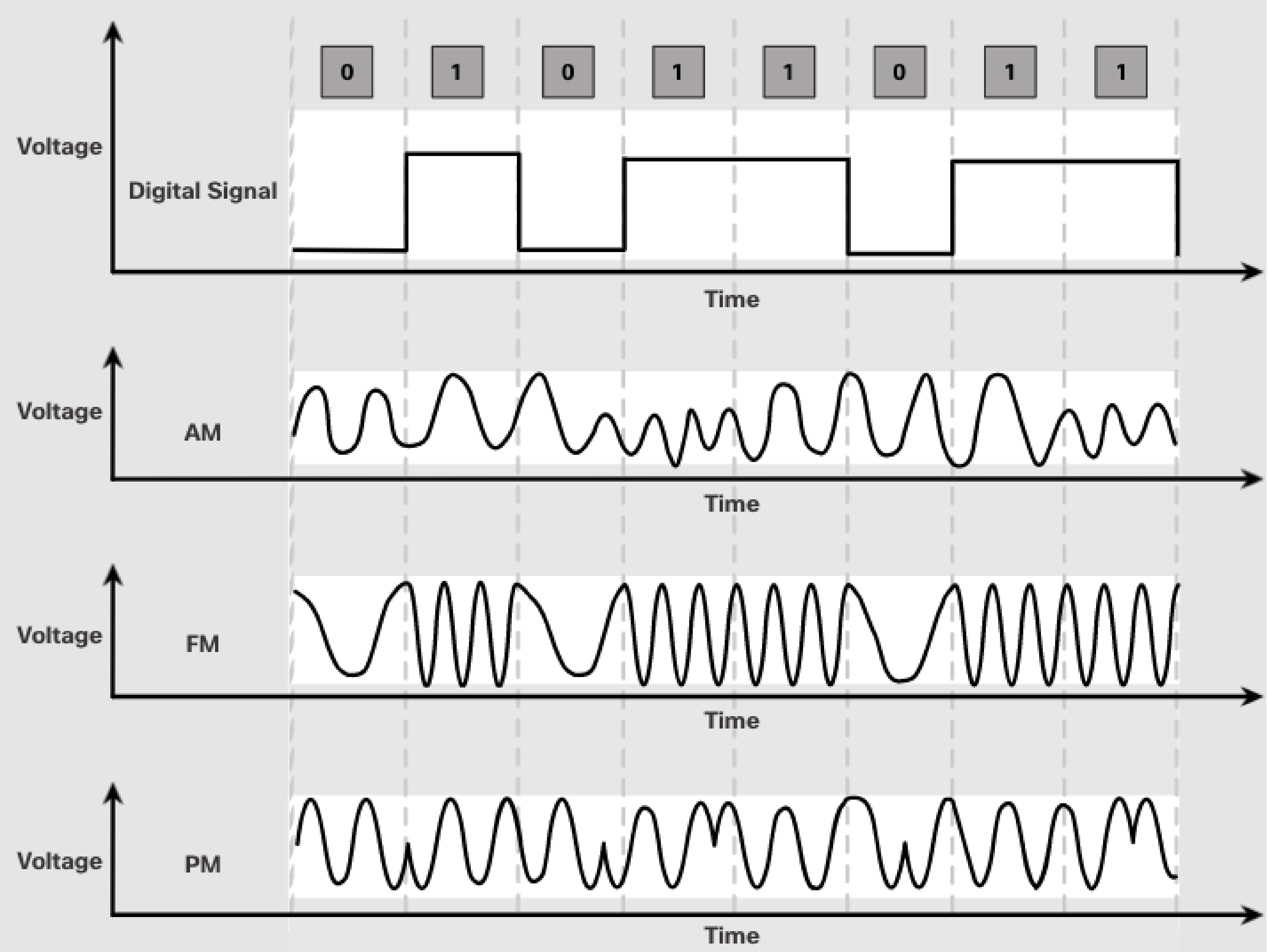
# Data Link Layer




Ethernet sample:

| Preamble | Destination Address | Source Address | Len | DATA | CRC |
|---|---|---|---|---|---|
| 8 bytes | 6 | 6 | 2 | 0-1500 | 4 |

# Physical Layer

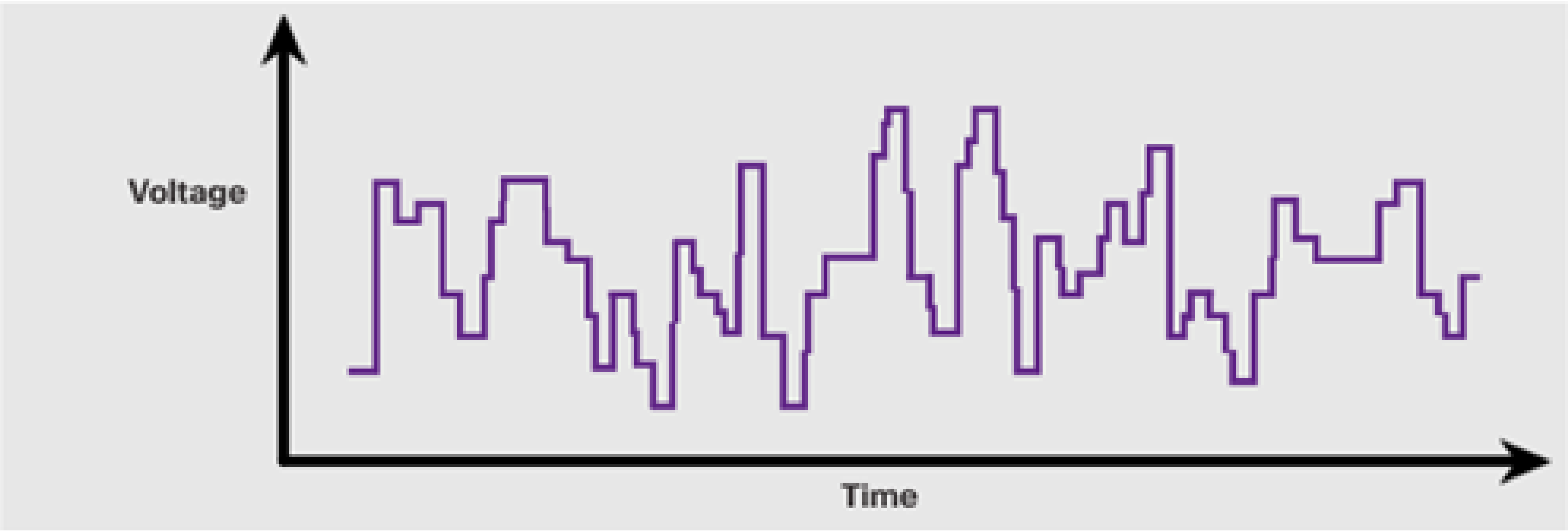



Determines the specs for all physical components

- Cabling
- Interconnect methods (topology / devices)
- Data encoding (bits to waves)
- Electrical properties

Examples:

- Ethernet (IEEE 802.3)
- Token Ring (IEEE 802.5)
- Wireless (IEEE 802.11)



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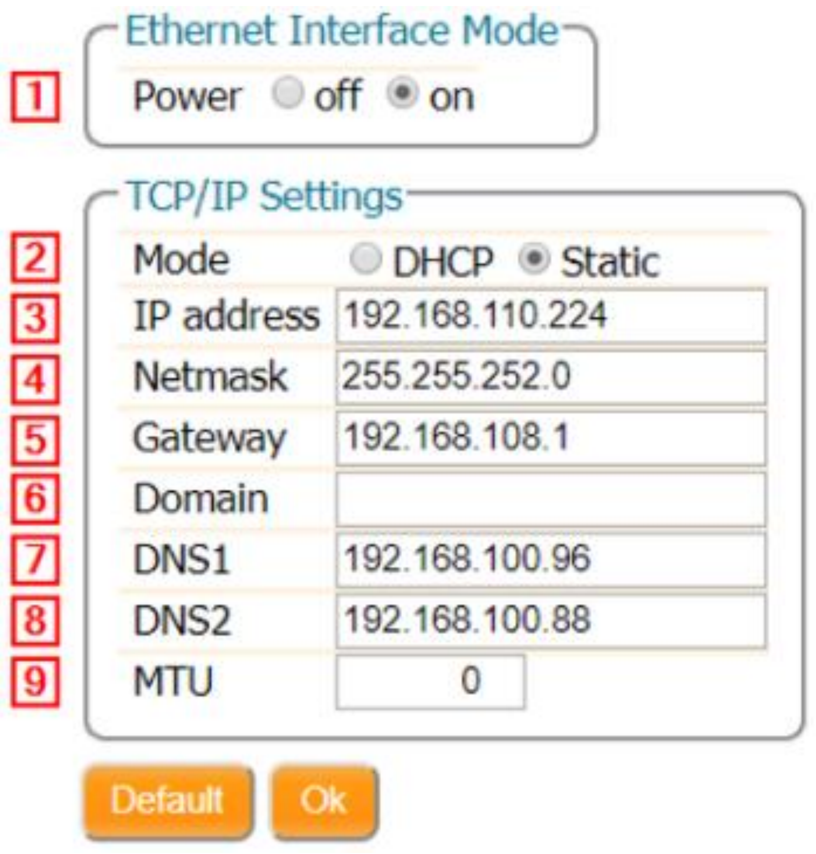
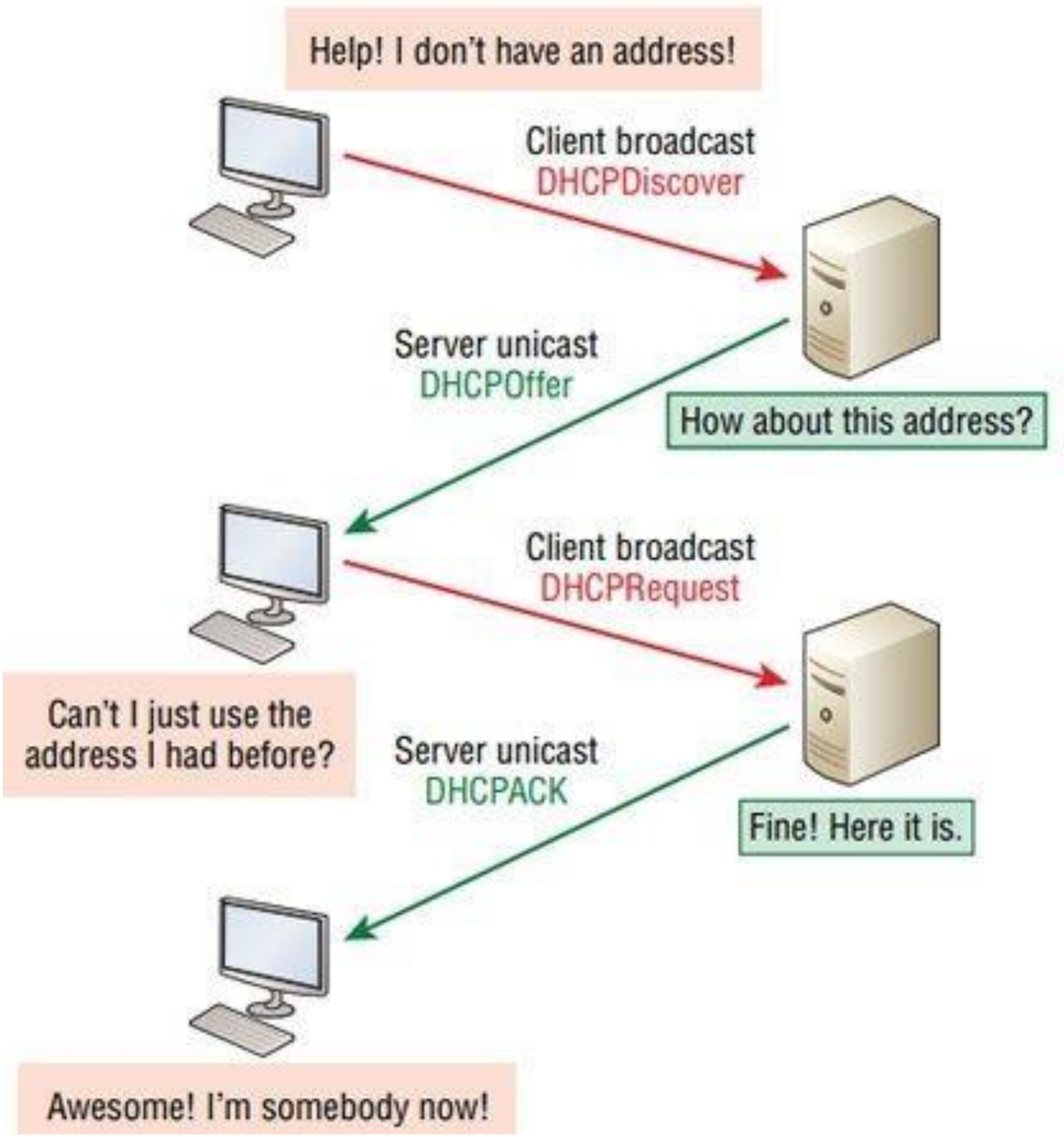
GET YOU'RE HUNTING EQUIPMENT, WE'RE GOING DEEP INTO THE WOODS

# Network services and roles

# Static vs Dynamic IP Address




| | |
|---|---|
| A static IP means that the same IP address is used every time you connect. | A dynamic IP address means that the IP changes every time they connect. |
| Static IP's allow you to offer hosting, such as emails, or media or gaming servers.<br><br>Great for businesses that receive large amounts of data. | A dynamic IP address is less vulnerable to attacks as the address is periodically changing.<br><br>Dynamic IP's are lower cost. |
| Offers a more stable connection and reduced down time as the IP never resets.<br><br>Static IP's are vulnerable to attacks, so extra security measures should be taken. | The network configures automatically,<br><br>Cannot offer hosting services. |
| Need to be configured manually.<br><br>It's difficult to transfer server settings from a static IP to a new computer | If the network fails then all the computers will try and find an IP address or lose their IP address completely. |

# Domain Name System

## DNS Server types

**By hierarch structure***:*

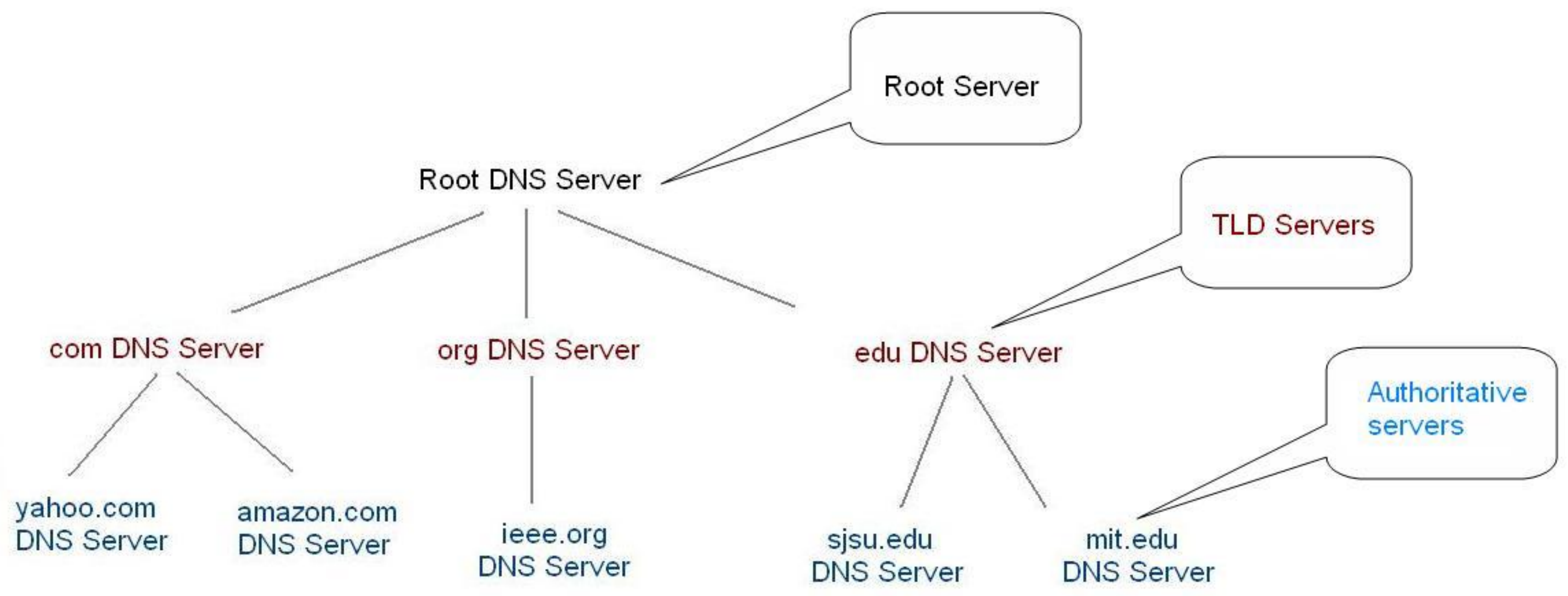
- *Root Servers*
- *TLD Servers*
- *Authoritative*

**By operation mode***:*

- *Authoritative*
- *Resolvers*

**By config management:**

- *Master*
- *Slave*




| Top-Level Domain | General Purpose | New TLDs | General Purpose |
|---|---|---|---|
| .com | U.S. Commercial | .biz | Business |
| .net | Network | .info | Information |
| .org | Nonprofit organization | .pro | Professional |
| .edu | U.S. Educational | .museum | Museums |
| .int | International | .aero | Aerospace industry |
| .mil | U.S. Military | .coop | Cooperative |
| .gov | U.S. Government | | |

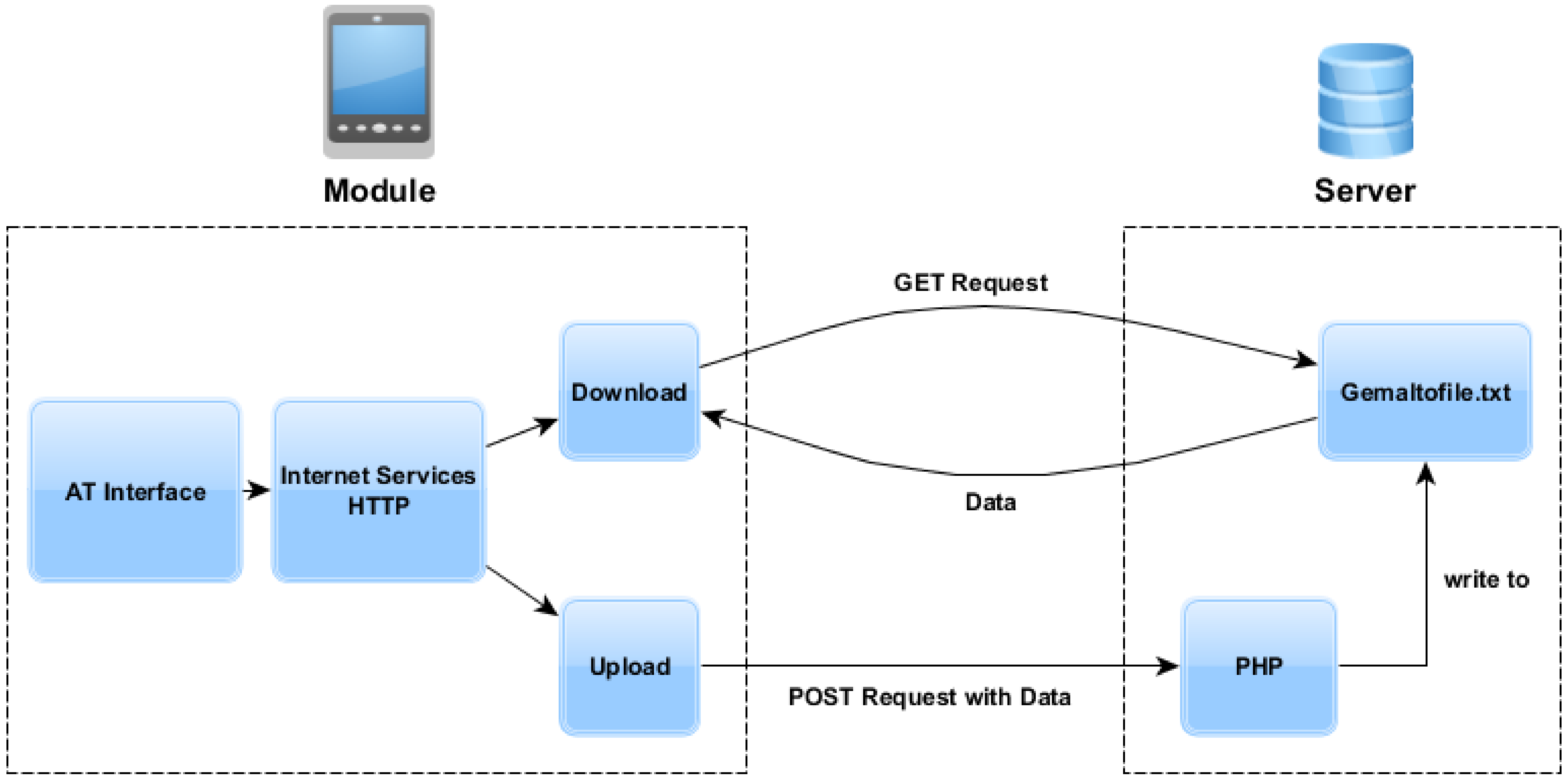
| Country Code TLD | Country |
|---|---|
| .au | Australia |
| .br | Brazil |
| .ca | Canada |
| .gr | Greece |
| .in | India |
| .ru | Russian Federation |
| .uk | United Kingdom |

## DNS Record types

- A (Host address)
- NS (Name Server)
- SOA (Start Of Authority)
- AAAA (IPv6 host address)
- CNAME (Canonical name for an alias)
- MX (Mail eXchange)

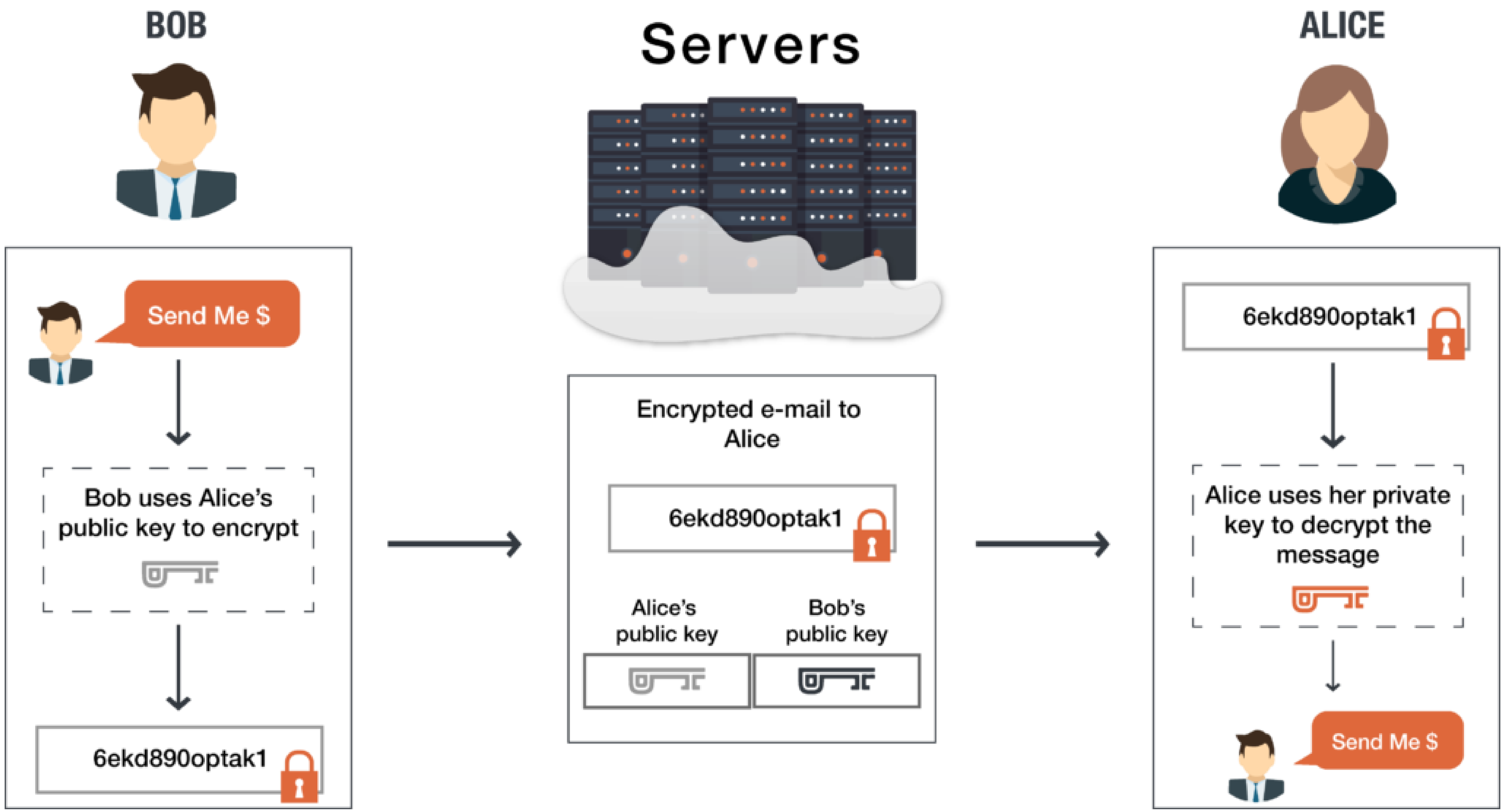
# Web services

HTTP versions:
- 1.0
- 1.1
- 2.0
- ….




**HTTP request**
- ***Schema:*** *HTTP, HTTPS*
- ***Method:*** *GET, POST, PUT, DELETE*
- ***Headers:*** *authentication, cookie support*
- **Payload:** XML, JSON

# Public, private keys



# Public, private keys (SSH)

| | |
|---|---|
| **RSA** | an old algorithm based on the difficulty of factoring large numbers. A key size of at least 2048 bits is recommended for RSA; 4096 bits is better. RSA is getting old and significant advances are being made in factoring. Choosing a different algorithm may be advisable. It is quite possible the RSA algorithm will become practically breakable in the foreseeable future. All SSH clients support this algorithm. |
| **DSA** | an old US government Digital Signature Algorithm. It is based on the difficulty of computing discrete logarithms. A key size of 1024 would normally be used with it. DSA in its original form is no longer recommended. |
| **ECDSA** | a new Digital Signature Algorithm standarized by the US government, using elliptic curves. This is probably a good algorithm for current applications. Only three key sizes are supported: 256, 384, and 521 (sic!) bits. We would recommend always using it with 521 bits, since the keys are still small and probably more secure than the smaller keys (even though they should be safe as well). Most SSH clients now support this algorithm. |
| **ECDSA** | this is a new algorithm added in OpenSSH. Support for it in clients is not yet universal. Thus, its use in general purpose applications may not yet be advisable. |

**CREATE A KEY**

ssh-keygen -t rsa -b 4096 ssh-keygen -t dsa ssh-keygen -t ecdsa -b 521 ssh-keygen -t ed25519

**COPY A KEY**

sh-keygen -f ~/tatu-key-ecdsa -t ecdsa -b 521

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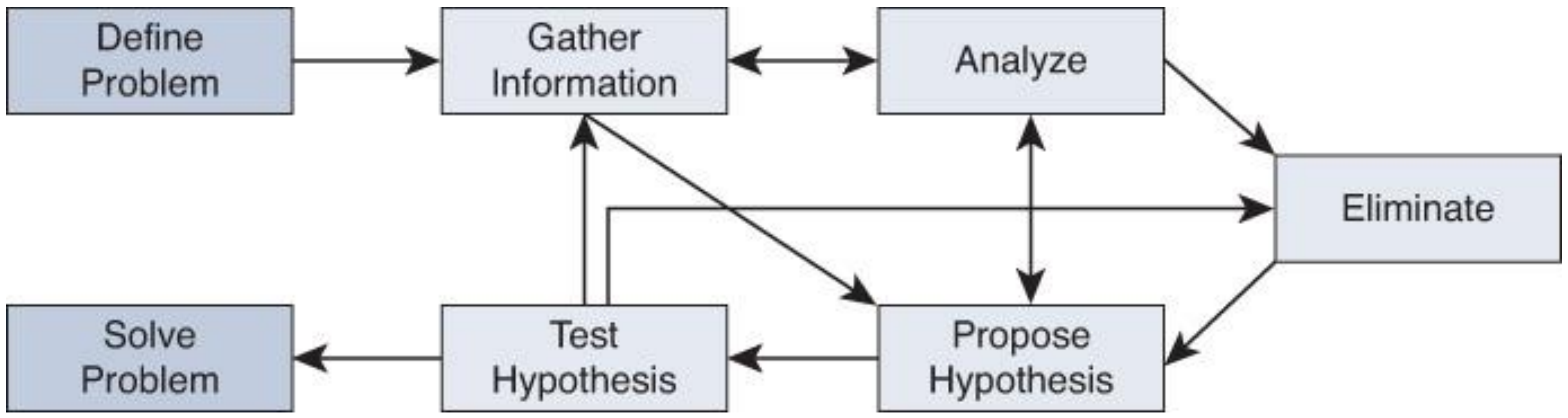
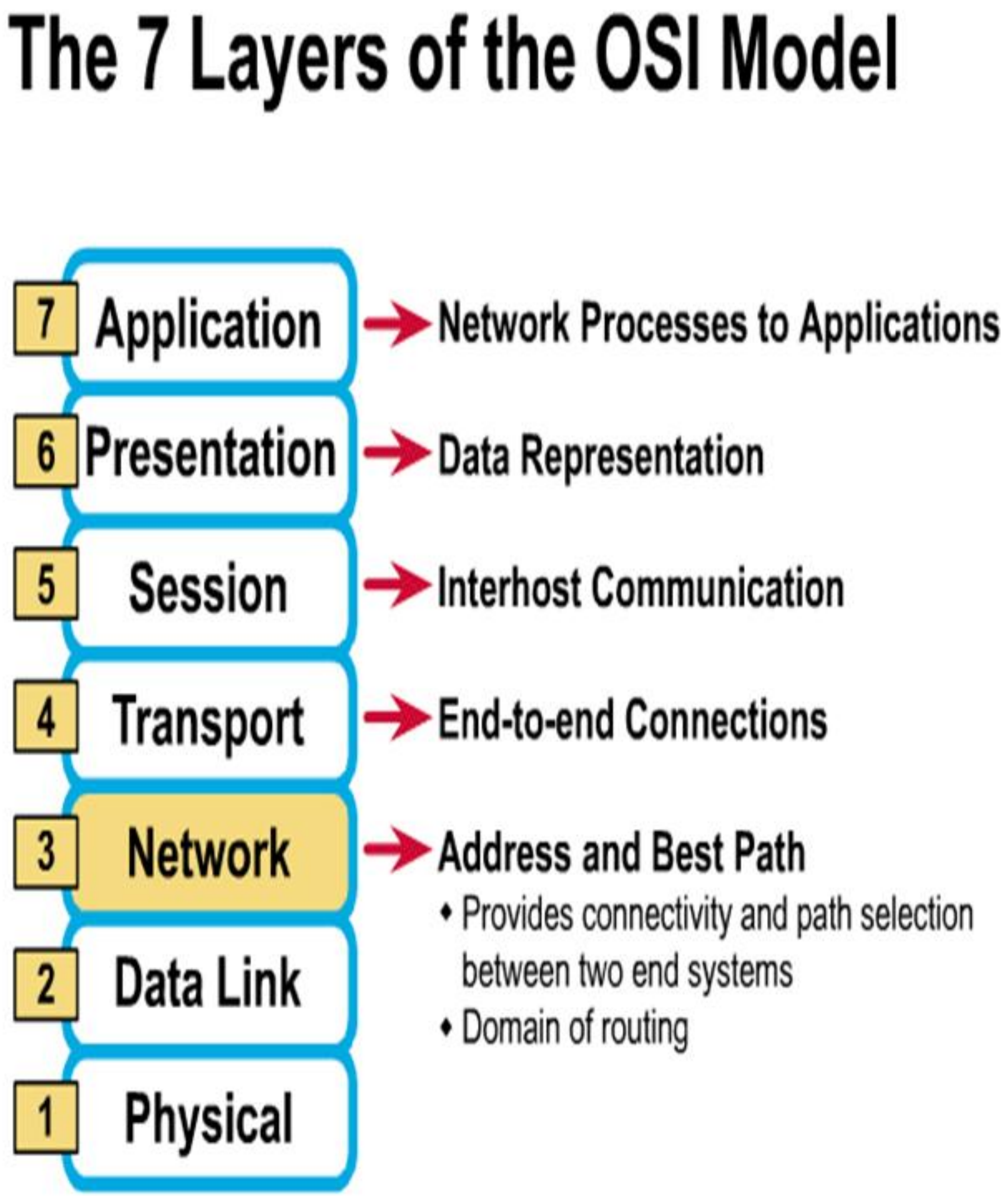
IF YOU'RE INTO FISHING THIS WILL BE VERRY USEFUL

# Troubleshooting approach and tools

- **Top-Down Approach** - look first at the application layer ***(OSI model)*** and then works down based on whether a problem has been found.
- **Bottom-Up Approach** - The Bottoms-Up approach uses the same OSI model as a basis but simply looks at the physical layers first
- **Divide and Conquer** - starts in the middle and works in the direction of the problem, problem is typically found faster than with the Top-Down or Bottoms-Up methods
- **Follow the Path - f**ollowing the path that the traffic takes through the network.
- **Spot the Differences -** when there is something(router, switch configurations) to compare against.



Configuration network tools

- *ifconfig, ifup, ifdown*
- *dhclient*
- *route*
- *ip addr*

## Commonly used configuration

- ip-address pool
- host-name
- domain-name
- subnet-mask
- routers
- domain-name-servers

Ubuntu/ Debian

- */etc/networking/interfaces*
- */etc/netplan/01-netcfg.yaml*
- *sudo netplan  apply*

Click to add text

Redhat/ Centos

- */etc/sysconfig/network-scripts/ifcfg-eth0*
- */etc/sysconfig/network-scripts/route-eth0*
- *nano /etc/sysconfig/network*

Windows

- *netsh*



**NETWORK**

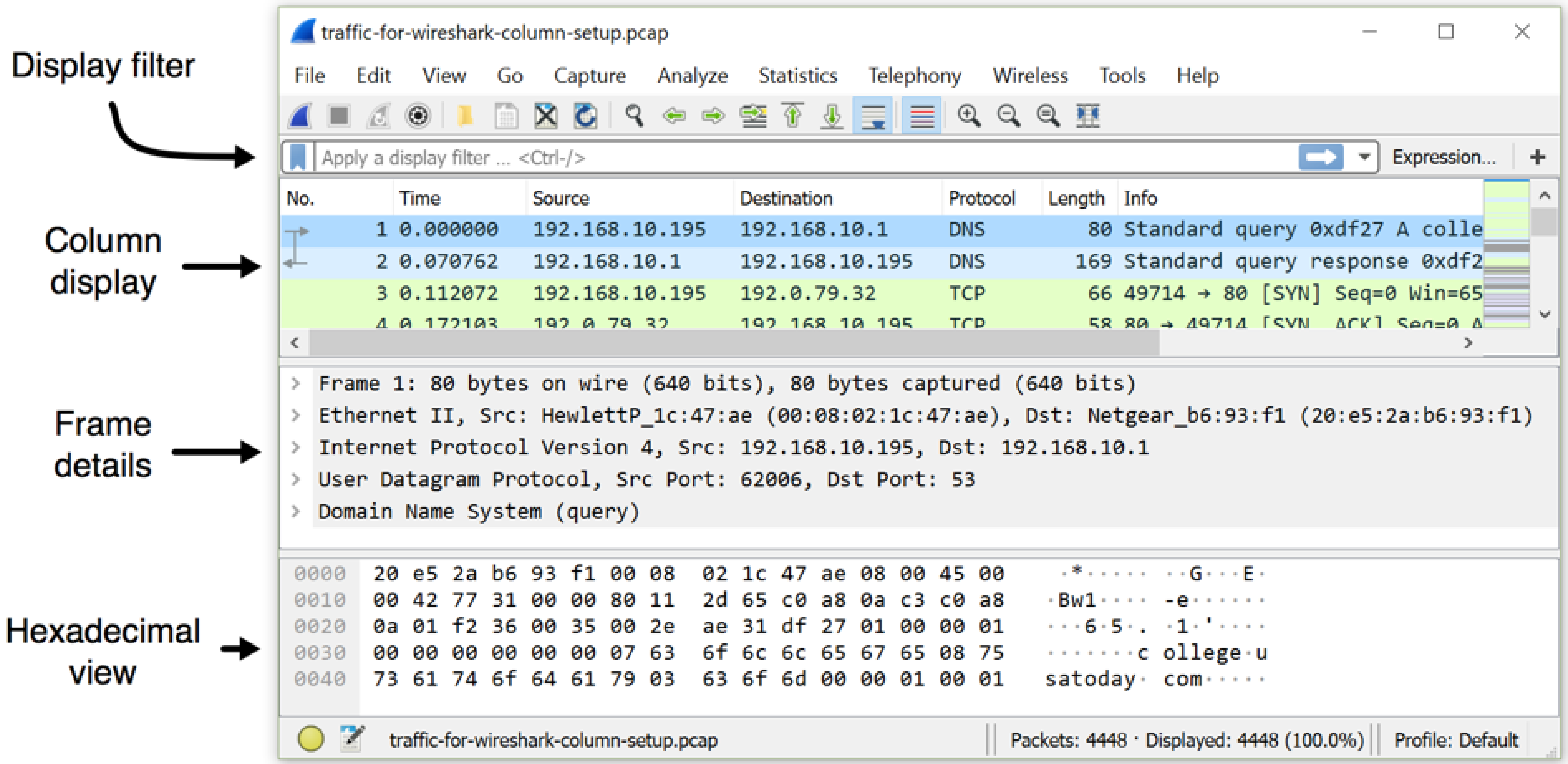
- ip, ifconfig, route
- ping
- traceroute, tracepath, tcptraceroute
- mtr
- netstat, ss
- nc, nmap
- tcpdump, wireshark
- arpwatch

**DNS**

- nslookup
- dig
- whois
- host

**WWW**

- wget
- curl
- openssl
- ssldump


Display filter
Column display
Frame details
Hexadecimal view
traffic-for-wireshark-column-setup.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Expression...
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.10.195 192.168.10.1 DNS 80 Standard query 0xdf27 A colle
2 0.070762 192.168.10.1 192.168.10.195 DNS 169 Standard query response 0xdf2
3 0.112072 192.168.10.195 192.0.79.32 TCP 66 49714 → 80 [SYN] Seq=0 Win=65
Frame 1: 80 bytes on wire (640 bits), 80 bytes captured (640 bits)
Ethernet II, Src: HewlettP_1c:47:ae (00:08:02:1c:47:ae), Dst: Netgear_b6:93:f1 (20:e5:2a:b6:93:f1)
Internet Protocol Version 4, Src: 192.168.10.195, Dst: 192.168.10.1
User Datagram Protocol, Src Port: 62006, Dst Port: 53
Domain Name System (query)
0000 20 e5 2a b6 93 f1 00 08 02 1c 47 ae 08 00 45 00
0010 00 42 77 31 00 00 80 11 2d 65 c0 a8 0a c3 c0 a8
0020 0a 01 f2 36 00 35 00 2e ae 31 df 27 01 00 00 01
0030 00 00 00 00 00 00 07 63 6f 6c 6c 65 67 65 08 75
0040 73 61 74 6f 64 61 79 03 63 6f 6d 00 00 01 00 01
traffic-for-wireshark-column-setup.pcap
Packets: 4448 · Displayed: 4448 (100.0%)
Profile: Default

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FOUND THESE LYING AROUND THE HOUSE SO I THOUGHT YOU MIGHT FIND THEM USEFUL

# Resources

- The Dawn of the Net
- Practical network flow
- IP Addressing and Subnetting for New Users
- Introduction to Packet Tracer
- How DHCP works
- Mobaxterm (enhance terminal tool)
- Wireshark portable
- TCP ping
- Sysinternals tools
- Linux dummy interface
- Linux IP Forward
- Homework -> IP fundamentals task.txt